

Indian Institute of Information Technology Guwahati
 CS306 - Machine Learning, Mid-Semester Examination
 Monsoon Semester AY 2025-26, Full Marks: 40, Time: 2 Hours

80.5

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Write the answer to PART A in the following table by filling in the circles.

		Questions														
		Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15
Options	A	☒	☒	☒	☒	☐	☐	☐	☐	☐	☒	☐	☐	☐	☒	☐
	B	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☒	☒	☐	☐	☐
	C	☐	☐	☐	☒	☐	☐	☒	☒	☒	☐	☐	☒	☒	☐	☒
	D	☐	☐	☐	☐	☐	☐	☐	☒	☒	☐	☐	☐	☒	☐	☐

+1 +1 0 +1 0 0 0.5 0.5 +1 0.5 0.5 +1 +1 0.5 0.5
 -1

PART A
 (Total Marks: 15)

Instructions for PART A: Each question has at least one correct answer. You score (i) 1 if you select all correct answers, (ii) 0.5 if you select at least one correct answer and no wrong answer, (iii) 0 if you do not answer, and (iv) -1 if you select at least one wrong answer.

- Suppose two variables X and Y have a Pearson correlation coefficient of $r = 0$. Which of the following is the most correct interpretation?
 - A. There is no linear relationship between X and Y .
 - B. The regression line of Y on X will be horizontal.
 - C. There is no relationship between X and Y .
 - D. X and Y are independent.
- Which of the following scenarios will **not** change the value of the Pearson correlation coefficient between X and Y ?
 - A. Adding a constant to all values of Y
 - B. Scaling all values of X by a positive constant
 - C. Adding random noise to Y that is uncorrelated with X
 - D. Multiplying all values of X by -1
- Suppose $\text{Cov}(X, Y) = 12$, $\sigma_X = 4$, and $\sigma_Y = 6$. What is the Pearson correlation coefficient r_{XY} ?
 - A. 2.00
 - B. 3.00
 - C. 0.25
 - D. 0.50
- Which of the following statements about Euclidean distance are correct?
 - A. It is the most common distance metric in K-means
 - B. It is less sensitive to outliers than Manhattan distance
 - C. It assumes features are on the same scale
 - D. It accounts for correlation among features
- Mahalanobis distance is useful because:
 - A. It ignores feature scaling
 - B. It accounts for correlations between variables
 - C. It uses the inverse covariance matrix
 - D. It is equivalent to Manhattan distance
- For ordinal data, rank-based distances:
 - A. Always yield zero distance
 - B. Often require normalization
 - C. Preserve order of categories
 - D. Ignore ordering of categories
- Hamming distance is suitable for:
 - A. Highly correlated multivariate data
 - B. Continuous numerical data
 - C. DNA sequence comparison
 - D. Binary data
- Jaccard distance between two sets is:
 - A. $\frac{|X \cap Y|}{|X \cup Y|}$
 - B. Always equal to Hamming distance
 - C. $1 - \frac{|X \cap Y|}{|X \cup Y|}$
 - D. Useful for sparse binary data
- In simple linear regression $y = w_0 + w_1x + \epsilon$:
 - A. ϵ is the independent variable
 - B. x is the dependent variable
 - C. w_0 is the intercept
 - D. w_1 is the slope
- The ordinary least squares (OLS) method:
 - A. Minimizes the sum of squared errors
 - B. Ignores variance of residuals
 - C. Provides a closed-form solution
 - D. Always requires gradient descent
- R^2 (coefficient of determination):
 - A. Can be negative in standard definition
 - B. Lies between 0 and 1
 - C. Measures variance explained by the model
 - D. Always equals correlation coefficient
- Gradient descent in regression:
 - A. Guarantees global optimum for non-convex functions
 - B. Sensitive to learning rate
 - C. Has variants: batch, stochastic, mini-batch
 - D. Updates parameters in direction of negative gradient
- Which are the hyperparameters in linear regression training?
 - A. w_0, w_1
 - B. Residuals
 - C. Number of epochs
 - D. Learning rate
- Overfitting occurs when:
 - A. Model performs well on training but poorly on test data
 - B. Model underfits both training and test data
 - C. Model has low training accuracy
 - D. Model learns noise in the dataset
- In multiple linear regression, the closed form solution is valid if:
 - A. Independent variables are perfectly correlated
 - B. Dataset size is always large
 - C. No severe multicollinearity exists
 - D. $X^T X$ is invertible